2019-Summer Paper 4 – Recent Advances

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Q1. Discuss neuromodulation for voiding dysfunction.

Type: LAQ (25 marks)

Definition and history

- **Neuromodulation:** electrical stimulation of a peripheral nerve that alters the activity of central pathways controlling the bladder. It acts on the nervous system, not directly on muscle.
- Direct muscle stimulation failed (painful, local). Stimulating **afferents** needs less current and affects the whole organ.
- **Tanagho and Schmidt** (about 1985): SNM (sacral neuromodulation). PTNS (percutaneous tibial nerve stimulation): FDA (Food and Drug Administration) approval 2000.
- **Rule:** exclude anatomical causes first (obstruction, stones, cancer, fistula). Stimulation cannot correct anatomy or restore muscle.

Mechanism

- Sacral roots are mixed nerves. Current excites **afferents → spinal cord and forebrain** (areas of awareness and attention) → altered PMC (pontine micturition centre) output.
- **OAB (overactive bladder):** heightened central control damps PMC firing → fewer involuntary contractions.
- **Non-obstructive retention:** restores awareness of filling and pelvic floor tone → PMC switches on → **EUS (external urethral sphincter) relaxes**. Hence SNM treats opposite disorders.
- **Tibial:** nerve carries **S2–S3** fibres; inhibition of OAB-type (C-fibre) reflexes is **opioid-mediated** (naloxone blocks it); long-term cortical plasticity.

Modalities

Modality	Target and method	Main use
SNM	Tined lead in **S3** foramen + implanted generator	Refractory OAB, NOUR (non-obstructive urinary retention)
PTNS	34-gauge needle above medial malleolus	Refractory OAB
TTNS (transcutaneous tibial)	Surface patch, home use	OAB (less effective)
Implantable tibial	Subcutaneous device over tibial nerve	OAB, avoids clinic visits

Modality	Target and method	Main use
Pudendal	Tined lead in Alcock's canal	IDO (idiopathic detrusor overactivity), NDO (neurogenic DO), SNM failure
DGNS (dorsal genital nerve stimulation)	Dorsal penile/clitoral nerve	Experimental; NDO
IVES (intravesical electrical stimulation)	Intravesical cathode	Underactive bladder in children
SARS (sacral anterior root stimulator)	Anterior roots + dorsal rhizotomy	Complete SCI (spinal cord injury)

A. Sacral neuromodulation

- **Third line** for OAB (AUA (American Urological Association) and EAU (European Association of Urology) agree). **FDA approved:** UUI (urgency urinary incontinence) 1997; urgency-frequency and NOUR 1999. **Not approved:** bladder pain, pelvic pain, neurogenic bladder.
- **Test phase** (bladder diary): **PNE (percutaneous nerve evaluation)**: office, temporary lead, 5–7 days; or **staged implant**: permanent tined lead under fluoroscopy, test up to 4 weeks, then generator in the buttock (preferred for NOUR).
- **Landmark:** S3 about 9 cm above the coccyx tip, 1–2 cm lateral. **S3 response: perineal "bellows" + great-toe flexion** (S2: calf/foot rotation; S4: bellows only).
- **Implant if ≥50% improvement** in a diary parameter (retention: ≥50% fewer catheterisations).
- **Evidence: InSite**: success 61% vs 42% with medical therapy. **ROSETTA**: onabotulinumtoxinA 200 U reduced UUI slightly more (−3.9 vs −3.3 episodes/day) but caused more UTI (urinary tract infection) (24% vs 10%) and CIC (clean intermittent catheterisation). NOUR: most stop catheterising.
- **Problems:** pain, lead migration, loss of effect; **revision >30% at 5 years**; costly.
- **Advances:** rechargeable (≥15 years) devices; full-body **MRI (magnetic resonance imaging)-conditional**.

B. Tibial nerve stimulation

- **PTNS technique:** 34-gauge needle **3–4 cm (three fingerbreadths) above the medial malleolus**, behind the tibia; ground pad on same leg; 20 Hz, 0–10 mA. Confirm by **toe flexion/fanning** or sole tingling. **30 min weekly × 12 weeks**, then maintenance.
- **SUmiT** (n = 220, validated sham): response **55% vs 21%**. Extension: 77% relapse-free at 3 years with tapering maintenance.
- **OrBIT:** PTNS comparable to extended-release tolterodine, with fewer side effects.
- **TTNS:** home self-treatment; skin impedance limits amplitude, so less effective.
- **Implantable devices** remove clinic visits: **BlueWind** (miniature, external control; benefit sustained at 3 years); **eCoin** (coin-sized, battery-powered, office insertion under local anaesthesia).

C. Other targets

- **Pudendal nerve stimulation:** tined lead via posterior approach. Better than SNM in a blinded crossover; **salvage after failed SNM**. Off-label.
- **DGNS:** purely sensory; acute DO suppression in SCI; **conditional (on-demand)** stimulation possible. Electrode fixation is difficult.
- **IVES (Katona):** intravesical cathode in saline activates Aδ afferents → detrusor contraction. Needs some afferent and muscle function; daily sessions; the sham-controlled trial was negative.

- **Brindley SARS:** stimulates anterior S2–S4 roots → voiding, defecation, erection. **Dorsal rhizotomy** abolishes reflex incontinence and DSD (detrusor–sphincter dyssynergia). Sphincter relaxes faster than detrusor between bursts → **post-stimulus voiding**. Needs intact efferents.

Comparison (third-line OAB)

	PTNS	OnabotulinumtoxinA 100 U	SNM
Invasiveness	Lowest	Moderate	Implant
Drawback	12 weekly visits	UTI, retention/CIC	Revision, cost
Retention	No role	Worsens	**Treats**

- **Also know:** intravaginal stimulation (SUI, stress urinary incontinence); sacral dermatome stimulation; bilateral SNM no better; closed-loop SNM.

Diagram to draw: S3 lead with buttock generator, and PTNS needle above the medial malleolus.

> **Key points to remember:**
> - Afferent modulation explains SNM benefit in both OAB and retention.
> - S3: bellows + great-toe flexion; implant if ≥50% improvement.
> - SUmiT: PTNS 55% vs sham 21%; 12 weekly sessions.
> - ROSETTA: Botox 200 U slightly better for UUI, more UTI and CIC.

Q2. Discuss bladder preservation surgery for TCC of urinary bladder.

Type: LAQ (25 marks)

Read as: bladder-preserving treatment of muscle-invasive TCC (transitional cell carcinoma, urothelial carcinoma), including radical TURBT (transurethral resection of bladder tumour), partial cystectomy and trimodality therapy.

Introduction

- Standard for MIBC (muscle-invasive bladder cancer): cisplatin NAC (neoadjuvant chemotherapy) + RC (radical cystectomy) with PLND (pelvic lymph node dissection).
- RC is morbid and patients are often elderly. Preservation suits **(1) unfit patients** and **(2) fit, motivated patients with limited disease**.
- **Why single modalities fail:** they leave either the remaining urothelium or occult nodes untreated (cT2 understaged to pT3 in up to about 40%; occult nodes about 25%).

Options

Option	Status
Radical TURBT alone	Not a primary treatment
Partial cystectomy + PLND	Highly selected solitary tumours; urachal cancer
RT (radiotherapy) alone	Inferior; quasi-palliative
Chemotherapy alone	Not recommended
TMT (trimodality therapy)	**Best-studied, guideline-accepted**

A. Radical TURBT

- Rationale: about 10% are pT0 at cystectomy. Only for tumour limited to superficial muscle, with negative re-resection.
- About a third progress to MIBC on long follow-up; selected series report 5-year CSS (cancer-specific survival) about 82%.

B. Partial cystectomy

Ideal candidate:

- Small (**<3 cm**), **solitary** tumour in a mobile part of bladder (dome).
- **No CIS (carcinoma in situ)**, no hydronephrosis.
- 1–2 cm margin achievable away from the ureteric orifice and trigone; adequate residual capacity.
- **Treatment of choice for urachal adenocarcinoma** (en bloc dome, urachus and umbilicus).

Contraindications: CIS, multifocal tumour, trigone/bladder neck tumour, small bladder, prostatic urethral involvement.

Technique: NAC if cisplatin-eligible → PLND → cystotomy away from tumour → full-thickness excision with perivesical fat, **frozen-section margins** → reimplant ureter if needed → watertight closure → irrigation (prevents seeding) → drain; cystogram before catheter removal.

Outcomes: 5-year OS (overall survival) 67–70%, intact bladder 65–74%; salvage RC up to a quarter; late recurrence → lifelong follow-up.

C. Trimodality therapy

Components: (1) **maximal, visibly complete TURBT**; (2) **RT** to bladder ± pelvic nodes; (3) concurrent **radiosensitising chemotherapy**.

Selection criteria:

- cT2 (selected cT3–4a), small solitary tumour.
- **No hydronephrosis, no extensive CIS**, visibly complete TURBT.
- Good bladder function; motivated, compliant patient.
- Only 6–19% of MIBC meet strict criteria.

Radiosensitising regimens:

Regimen	Evidence
Mitomycin C + 5-FU (fluorouracil)	**BC2001**: 2-year locoregional DFS (disease-free survival) **67% vs 48%** vs RT alone, no excess late toxicity
Cisplatin (± 5-FU or paclitaxel)	MGH (Massachusetts General Hospital)/RTOG (Radiation Therapy Oncology Group) protocols
Gemcitabine	Cisplatin-ineligible
Carbogen + nicotinamide (BCON)	Hypoxia modification

Radiation schedules:

```
Split course (MGH/RTOG):  TURBT → chemoRT ~40 Gy → cystoscopy + biopsy
                           CR → consolidation to 60–66 Gy
                           Residual tumour → immediate cystectomy
Continuous (current):      TURBT → chemoRT 64 Gy/32 fr or 55 Gy/20 fr
                           → cystoscopy and imaging at 3–4 months
```

Response assessment: cystoscopy with biopsy of tumour site, cytology, CT (computed tomography)/MRI (magnetic resonance imaging). **CR (complete response)** → surveillance. Non-response → salvage RC.

Salvage cystectomy:

- Needed in **10–30%** for persistent or recurrent muscle-invasive disease; it is the designed safety net.
- NMIBC (non-muscle-invasive) recurrence → TURBT + intravesical BCG (bacillus Calmette–Guérin).

Outcomes:

- 5-year CSS 50–84%, OS 36–74%.
- MGH/RTOG: CR 69%; 5-year DSS (disease-specific survival) 71%; 80% of survivors keep the bladder.
- Propensity-matched RC vs TMT: equal 5-year metastasis-free survival (about 75%) and CSS (81% vs 84%).
- Low late toxicity; good quality of life.

Comparison with radical cystectomy

	Radical cystectomy	TMT
Staging	Pathological	Clinical only
Nodes	Removed	Irradiated or not treated

	Radical cystectomy	TMT
Function	Diversion; sexual dysfunction	Native bladder; bowel/bladder toxicity
Follow-up	Upper tract, urethra	**Intensive cystoscopy for 10 years**
Evidence	Standard	No completed RCT (SPARE closed for poor accrual)

Follow-up after preservation

- Cystoscopy every 3–4 months for 2 years, 6-monthly years 3–4, then yearly to 10 years; CT urogram and chest CT every 3–6 months for 2 years, then yearly to 5 years; cytology.

Recent advances

- **Immunotherapy + chemoradiation:** PD-1/PD-L1 (programmed death-1 / ligand-1) inhibitors with TMT under trial (e.g. SWOG/NRG S1806 with atezolizumab; KEYNOTE-992 with pembrolizumab); results awaited.
- Hypofractionation 55 Gy/20 non-inferior (BC2001–BCON meta-analysis: same toxicity).
- Bladder sparing after complete response to NAC (± checkpoint inhibitor) with biomarker selection (DNA repair gene mutations, circulating tumour DNA); RETAIN: about half recurred.
- **Also know:** robotic partial cystectomy; molecular subtypes.

Diagram to draw: TMT flowchart from TURBT to chemoradiation, response check, surveillance or salvage cystectomy.

> **Key points to remember:**
>
> - TMT = complete TURBT + RT + radiosensitiser; select T2, no hydronephrosis, no CIS.
> - BC2001: MMC/5-FU + RT, 2-year locoregional DFS 67% vs 48%.
> - Salvage cystectomy 10–30%; follow up the bladder for 10 years.
> - Partial cystectomy: solitary <3 cm, no CIS, frozen-section margins, always PLND.

Q3a. PSMA PET Ct scan

Type: SAQ (10 marks)

Principle

- **PSMA (prostate-specific membrane antigen):** transmembrane glycoprotein over-expressed by prostate cancer, rising with grade.
- **PET/CT (positron emission tomography/computed tomography):** positron-labelled ligand binds PSMA; uptake as SUV (standardised uptake value).
- **Tracers:** ^{68}Ga-PSMA-11 (half-life 68 min); ^{18}F-DCFPyL, ^{18}F-rhPSMA-7.3, ^{18}F-PSMA-1007 (110 min, better images; PSMA-1007 little urinary excretion).
- **Specificity >90%**; beats choline and fluciclovine.

Indications

1. Primary staging (high risk):

- **proPSMA (2020):** accuracy **92% vs 65%** vs CT + bone scan.
- Nodal sensitivity only 30–40% against PLND (pelvic lymph node dissection): a negative scan does not exclude micrometastases.
- NCCN (National Comprehensive Cancer Network): may **replace** conventional imaging; AUA (American Urological Association): add-on.

2. BCR (biochemical recurrence):

- Detection rises with PSA (prostate-specific antigen): about **half positive below 0.5 ng/mL**, nearly all above 2.
- **Calais trial:** PSMA 56% vs fluciclovine 26%. **CONDOR:** management changed in 63.9%.
- Guides salvage radiotherapy (disease outside field in about 1 in 5).

3. Intraprostatic detection: PRIMARY trial: adding PSMA PET to MRI (magnetic resonance imaging) improved negative predictive value; 5-point PRIMARY score.

4. Theranostics: selects patients for **^{177}Lu-PSMA-617**. **VISION (2021):** mCRPC (metastatic castration-resistant prostate cancer) after ARPI (androgen receptor pathway inhibitor) and taxane: OS 15.3 vs 11.3 months.

Reporting

- **PROMISE miTNM** (molecular imaging TNM): miT, miN, miM staging plus expression score (vs blood pool, liver, parotid). PSMA-RADS also used.

Pitfalls

False positive	False negative
Coeliac/stellate ganglia, fractures, Paget's disease, fibrous dysplasia	Small nodes (<5 mm)
Other tumours (RCC (renal cell carcinoma), HCC (hepatocellular carcinoma))	PSMA-negative tumours (about 5–10%)
^{18}F-PSMA-1007 benign bone uptake	Neuroendocrine transformation; masking by bladder activity

- **Also know:** no proof yet that PSMA-guided treatment improves survival (stage migration).

Diagram to draw: PSMA ligand binding the cell-surface antigen with fused PET/CT image of a nodal lesion.

Key points to remember:

- proPSMA: accuracy 92% vs 65%.
- BCR: about half positive below PSA 0.5 ng/mL.
- Nodal sensitivity 30–40%: negative scan does not exclude nodes.
- VISION: ^{177}Lu-PSMA-617 improves OS in mCRPC.

Q3b. LHRH agonist

Type: SAQ (10 marks)

Physiology

- **LHRH (luteinising hormone-releasing hormone, GnRH (gonadotrophin-releasing hormone))** from the hypothalamus is released in **pulses**. It drives pituitary LH (luteinising hormone) and FSH (follicle-stimulating hormone). LH → Leydig cells → testosterone (about 95% testicular).
- **Continuous** exposure **desensitises and down-regulates** the pituitary receptor. This is the basis of agonist therapy.

Drugs

Drug	Formulation
Leuprolide	Depot 1, 3, 6 monthly
Goserelin	SC (subcutaneous) implant 1 or 3 monthly
Triptorelin	IM (intramuscular) depot
Histrelin	Implant, 1 year
Buserelin	Nasal or SC

- Substitution at **position 6** of the decapeptide raises potency and half-life.

Mechanism

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Agonist → initial LH surge (flare, testosterone up)
        → receptor down-regulation (1–2 weeks)
        → LH falls → castrate testosterone in 2–4 weeks
```

- Survival **equals orchidectomy**. Castrate: **<50 ng/dL** (traditional); aim **<20 ng/dL** (orchidectomy median about 15).

Flare phenomenon

- LH rises up to **10-fold for 10–20 days** → bone pain, **spinal cord compression**, urinary retention.
- Dangerous with spinal or weight-bearing metastases.
- **Prevent:** antiandrogen (bicalutamide, flutamide) started with or before the agonist, for **21–28 days**.
- Impending cord compression → **antagonist** (degarelix) or orchidectomy instead.

Indications

- Metastatic hormone-sensitive prostate cancer (with ARPI (androgen receptor pathway inhibitor) or docetaxel).
- With radiotherapy for intermediate/high-risk and locally advanced disease.
- BCR (biochemical recurrence) with short doubling time; continued in CRPC (castration-resistant prostate cancer).
- **Also know:** central precocious puberty, endometriosis; cryptorchidism (low success); intermittent ADT (androgen deprivation therapy) acceptable (PR.7).

Side effects

- Hot flushes (50–80%), loss of libido, ED (erectile dysfunction).

- **Osteoporosis** (DEXA (dual-energy X-ray absorptiometry); calcium, vitamin D, denosumab or zoledronic acid).
- Diabetes, cardiovascular events, sarcopenia, anaemia, gynaecomastia (≤5%), cognitive change.
- Slow testosterone recovery after stopping.

Agonist vs antagonist

Agonist	Antagonist
Flare; needs antiandrogen	No flare; castrate by day 3 (degarelix)
FSH rises	FSH falls
Depot to yearly	Monthly SC or oral (relugolix; HERO)

Diagram to draw: hypothalamic–pituitary–testis axis showing the agonist site and flare.

> **Key points to remember:**
> - Continuous stimulation → down-regulation → castration in 2–4 weeks.
> - Flare 10–20 days; antiandrogen for 21–28 days.
> - Castrate <50 ng/dL; aim <20 ng/dL.

Q3c. Digital flexible ureteroscopes

Type: SAQ (10 marks)

Introduction

- **Fibreoptic flexible URS (ureteroscopy):** coherent glass-fibre bundle carries the image; fibres break → black spots; honeycomb (moiré) pattern.
- **Digital scopes** (first 2006): a sensor at the tip (**"chip-on-tip"**) converts light to an electrical signal, so the image does not degrade.

Sensor technology

	CCD (charge-coupled device)	CMOS (complementary metal-oxide semiconductor)
Heat and power	More	Less (runs cooler)
Cost, speed	Higher, slower	Cheaper, faster
Status	Phased out	Current standard

- **LED (light-emitting diode)** at the tip; no camera head or light cable → better ergonomics.

Fibreoptic vs digital

Feature	Fibreoptic	Digital
Image	Degrades	Sharper, magnified, stable
Tip	Smallest, tapered	Larger (about 8.5 Fr), flatter
Durability	Repair about every 15 cases	**Not better** in later comparisons
Operative time	Longer	Shorter; similar SFR (stone-free rate)
Processor	Platform-agnostic	Proprietary

- Larger tip → more access-sheath use. Keep a fibreoptic scope for narrow UPJ (pelviureteric junction) or infundibulum.

Deflection and channel

- Active deflection **≥270° both directions**; enhanced primary deflection eases lower-pole access.
- Working channel **3.6 Fr**; dual-channel scopes exist.

Single-use vs reusable

- **LithoVue (2016):** first single-use digital scope; CMOS, 7.7 Fr tip, 9.5 Fr shaft, 270°.
- Single-use advantages: maximal deflection each case, sterile, no reprocessing or repair; cost **lower below about 99 cases/year**.
- Reusable: cheaper in high-volume centres; needs reprocessing (peracetic acid, glutaraldehyde, ethylene oxide).
- Most centres use a **hybrid fleet**; environmental impact similar.

Recent advances

- **Intrarenal pressure sensor** in the scope (LithoVue Elite): high pressure causes fever, SIRS (systemic inflammatory response syndrome), sepsis. Temperature sensing next.
- **AI (artificial intelligence):** stone recognition, laser setting adjustment, navigation.
- **Robotic flexible URS:** ergonomics, less radiation to surgeon.

- **Also know:** suction sheaths, further miniaturisation.

Diagram to draw: scope tip cross-section showing CMOS chip, LED and 3.6 Fr channel.

> **Key points to remember:**
> - Chip-on-tip CMOS: better image, larger tip, durability not better.
> - Single-use cheaper below about 99 cases/year.
> - Pressure sensors and AI are the newest developments.

Q3d. Videoendoscopic inguinal lymphadenectomy

Type: SAQ (10 marks)

Definition

- **VEIL (video-endoscopic inguinal lymphadenectomy):** ILND (inguinal lymph node dissection) through thigh ports, beneath Scarpa's fascia; laparoscopic (Tobias-Machado) or robotic (**RVEIL**).
- **Aim:** open-surgery template without the groin incision that causes flap necrosis.

Indications

- Penile SCC (squamous cell carcinoma), **impalpable nodes**, intermediate/high risk (pT1b+, G3, LVI (lymphovascular invasion)).
- **Mobile, non-bulky palpable nodes.**
- **Not for** bulky, fixed (cN3) or skin-involving nodes → chemotherapy, then open surgery.
- Also vulval cancer, melanoma.

Boundaries (femoral triangle)

- Superior: **inguinal ligament**; lateral: **sartorius**; medial: **adductor longus**; floor: fascia lata; apex: muscles crossing.
- Mark an inverted triangle on the skin **before insufflation**, because gas distorts landmarks.

Technique

```
Position: supine frog-leg / low lithotomy
→ 2 cm camera incision below the femoral triangle apex
→ finger dissection beneath Scarpa's fascia
→ two lateral and medial working ports (+ assistant port)
→ CO2 at 15 mmHg (higher briefly to create space)
→ develop space up to the inguinal ligament
→ remove nodal packet off fascia lata, apex to ligament
→ preserve saphenous vein if possible; clip tributaries
→ frozen section → deep dissection medial to femoral vein if positive
→ bag retrieval, suction drain in dependent part
```

- **Template:** superficial ± deep; Daseler's five zones; Cloquet's node is the highest deep node.
- **Never dissect lateral to the femoral artery** (femoral nerve).

Outcomes vs open

	Open ILND	VEIL/RVEIL
Wound infection, necrosis, dehiscence	Higher	**Lower**
Hospital stay, drain time	Longer	Shorter
Operative time	Shorter	About 2 h (slightly longer per side)
Nodal yield, recurrence	Reference	**Similar**
Evidence	Retrospective	Retrospective, selection-biased

- Commonest complication: **lymphocele**, then lymphoedema and cellulitis.

Robotic VEIL

- 3D vision, wristed instruments, ICG (indocyanine green) for lymphatics; adequate dissection on open re-exploration (phase I). No VEIL vs RVEIL comparison.

- **Also know:** DSNB (dynamic sentinel node biopsy) first for impalpable nodes (EAU-ASCO, European Association of Urology–American Society of Clinical Oncology); modified (Catalona) template; complete nodal management within 3 months.

Diagram to draw: femoral triangle with port sites below the apex.

> **Key points to remember:**
> - Same template as open: ligament, sartorius, adductor longus.
> - Fewer wound complications; equal nodal yield.
> - Not for bulky, fixed or skin-involving nodes.

Q3e. Methods to predict prognosis of patient of Cancer prostate

Type: SAQ (10 marks)

1. Clinical and pathological factors

- **PSA (prostate-specific antigen):** organ-confined about 80% at <4 but <50% at >10 ng/mL; density and velocity add information.
- **Clinical stage (DRE (digital rectal examination), TNM).**
- **Grade group (ISUP (International Society of Urological Pathology)) – strongest single factor:**

Grade group	Gleason	5-year BCR (biochemical recurrence)-free
1	≤6	96%
2	3+4	88%
3	4+3	63%
4	8	48%
5	9–10	26%

- Biopsy: % positive cores, **cribriform and intraductal** patterns, perineural invasion.
- Prostatectomy: margin (extent, grade), EPE (extraprostatic extension), seminal vesicle and node status.

2. Risk groups

D'Amico	Low	Intermediate	High
PSA (ng/mL)	≤10	10–20	>20
Gleason	≤6	7	8–10
Stage	T1–T2a	T2b	T2c

- **NCCN (National Comprehensive Cancer Network):** very low, low, favourable/unfavourable intermediate, high, very high.
- **CAPRA (Cancer of the Prostate Risk Assessment)** score 0–10 (age, PSA, Gleason, stage, % cores); **CAPRA-S** after surgery.

3. Nomograms

- **Partin tables:** organ-confined disease, EPE, seminal vesicle, node involvement.
- **MSKCC (Memorial Sloan Kettering) nomograms:** pre- and post-treatment BCR and 15-year cancer-specific survival.
- **Briganti nomogram:** risk of nodal invasion → decides PLND (pelvic lymph node dissection).

4. Imaging

- mpMRI (multiparametric MRI): PI-RADS (Prostate Imaging Reporting and Data System) score, tumour contact length, EPE.
- PSMA PET (prostate-specific membrane antigen PET) for metastatic burden.

5. Tissue biomarkers

Test	Basis	Use
Decipher	22-RNA classifier	Metastasis risk; post-op radiotherapy
Oncotype DX GPS	17 genes	Adverse pathology; active surveillance

Test	Basis	Use
Prolaris	Cell-cycle progression genes	Progression, cancer death

- Germline BRCA2 (BReast CAncer gene 2): aggressive disease.

6. After treatment

- **PSA doubling time (PSADT):** <3 months highest risk; Pound: time to BCR <2 years, PSADT <10 months, Gleason 8–10 predict metastasis.
- **EAU (European Association of Urology) BCR risk groups:** low risk = PSADT >1 year and grade group <4.
- PSA nadir on ADT (androgen deprivation therapy); **Phoenix** failure (nadir + 2) after radiotherapy.

Diagram to draw: flowchart from risk group to nomogram, biomarker and treatment decision.

> **Key points to remember:**
> - Grade group is the strongest predictor.
> - D'Amico, NCCN and CAPRA combine PSA, grade and stage.
> - Decipher, Oncotype, Prolaris add genomic risk.

Q3f. Male menopause

Type: SAQ (10 marks)

Read as: late-onset hypogonadism (testosterone deficiency of ageing men, "andropause"). Same topic as 2017-Summer Paper 4, Q3e and 2017-Winter Paper 4, Q3e; full answer given here.

Definition

- Symptoms plus **consistently low serum testosterone** in an ageing man.
- "Menopause" is a misnomer: the fall is gradual and partial. Preferred terms: PADAM (partial androgen deficiency of the ageing male), LOH (late-onset hypogonadism).

Pathophysiology

- Testosterone falls slowly from middle age.
- **Testis:** fewer Leydig cells. **Centre:** blunted feedback, irregular GnRH (gonadotrophin-releasing hormone) pulses.
- **SHBG (sex hormone-binding globulin) rises** with age, so free testosterone falls more.
- Worse with obesity, diabetes and **opioids**.

Clinical features

- **Sexual (most specific):** low libido, ED (erectile dysfunction), fewer morning erections.
- Fatigue, low muscle mass, osteoporosis, anaemia, low mood.
- **ADAM (Androgen Deficiency in the Aging Male)** questionnaire: sensitive, not specific.

Diagnosis

- **Morning (7–11 am) total testosterone, twice.** AUA (American Urological Association): **<300 ng/dL**. EAU (European Association of Urology): about **12 nmol/L**; calculated free testosterone if borderline.
- **LH (luteinising hormone):** high in primary, low or normal in secondary hypogonadism. Check **prolactin** if LH is low; pituitary MRI (magnetic resonance imaging) if testosterone <150 ng/dL.
- Baseline: PSA (prostate-specific antigen), DRE (digital rectal examination), haematocrit.

Treatment

- Lifestyle first (weight loss, exercise).
- **TRT (testosterone replacement therapy)** for symptoms + confirmed low level: gels, injections (enanthate every 2–3 weeks; undecanoate every 10–14 weeks), oral undecanoate. Stop if no benefit at 3–6 months.
- **Contraindications:**
  - Prostate or breast cancer.
  - **Wish for fertility** (TRT stops sperm production; use hCG (human chorionic gonadotrophin) or clomiphene).
  - Haematocrit >50%.
  - Untreated severe sleep apnoea; recent MI (myocardial infarction).

Monitoring

- At 3–6 months, then yearly: testosterone, **haematocrit** (stop if >54%), PSA.

Safety evidence

- **TRAVERSE (2023):** 5,246 men aged 45–80 at high cardiovascular risk. TRT was **non-inferior for major cardiac events**; more atrial fibrillation and pulmonary embolism.

Diagram to draw: hypothalamic–pituitary–testis axis with ageing defects.

> **Key points to remember:**
>
> - Symptoms + low morning testosterone twice (AUA <300 ng/dL; EAU 12 nmol/L).
> - LH separates primary from secondary; prolactin if LH low.
> - TRT harms fertility; watch haematocrit; TRAVERSE: cardiac non-inferiority.

Rapid Revision Sheet – 2019-Summer Paper 4

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Discuss neuromodulation for voiding dysfunction. (LAQ)

- Afferent modulation explains SNM benefit in both OAB and retention.
- S3: bellows + great-toe flexion; implant if ≥50% improvement.
- SUmiT: PTNS 55% vs sham 21%; 12 weekly sessions.
- ROSETTA: Botox 200 U slightly better for UUI, more UTI and CIC.

Q2. Discuss bladder preservation surgery for TCC of urinary bladder. (LAQ)

- TMT = complete TURBT + RT + radiosensitiser; select T2, no hydronephrosis, no CIS.
- BC2001: MMC/5-FU + RT, 2-year locoregional DFS 67% vs 48%.
- Salvage cystectomy 10–30%; follow up the bladder for 10 years.
- Partial cystectomy: solitary <3 cm, no CIS, frozen-section margins, always PLND.

Q3a. PSMA PET Ct scan (SAQ)

- proPSMA: accuracy 92% vs 65%.
- BCR: about half positive below PSA 0.5 ng/mL.
- Nodal sensitivity 30–40%: negative scan does not exclude nodes.
- VISION: ^{177}Lu-PSMA-617 improves OS in mCRPC.

Q3b. LHRH agonist (SAQ)

- Continuous stimulation → down-regulation → castration in 2–4 weeks.
- Flare 10–20 days; antiandrogen for 21–28 days.
- Castrate <50 ng/dL; aim <20 ng/dL.

Q3c. Digital flexible ureteroscopes (SAQ)

- Chip-on-tip CMOS: better image, larger tip, durability not better.
- Single-use cheaper below about 99 cases/year.
- Pressure sensors and AI are the newest developments.

Q3d. Videoendoscopic inguinal lymphadenectomy (SAQ)

- Same template as open: ligament, sartorius, adductor longus.
- Fewer wound complications; equal nodal yield.
- Not for bulky, fixed or skin-involving nodes.

Q3e. Methods to predict prognosis of patient of Cancer prostate (SAQ)

- Grade group is the strongest predictor.
- D'Amico, NCCN and CAPRA combine PSA, grade and stage.
- Decipher, Oncotype, Prolaris add genomic risk.

Q3f. Male menopause (SAQ)

- Symptoms + low morning testosterone twice (AUA <300 ng/dL; EAU 12 nmol/L).
- LH separates primary from secondary; prolactin if LH low.
- TRT harms fertility; watch haematocrit; TRAVERSE: cardiac non-inferiority.